

Hari Priya P B

Software Engineer

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SUMMARY

Experienced and driven Software Engineer with 4+ years of building production systems across healthcare, e-pharmacy and enterprise domains. Hands-on expertise in microservices, RESTful APIs, event-driven architecture, workflow automation and system integration on AWS, with the ability to adapt quickly to new technology stacks and business contexts. Skilled in translating business requirements into business-critical applications, fostering cross-team collaboration, mentoring junior engineers and supporting systems in production. Enjoys simplifying complex, real-world problems and building technology that makes a meaningful difference to people. Further strengthened by an MSc in Advanced Computer Science at the University of Manchester, focusing on Machine Learning and AI, with hands-on projects in NLP, LLMs, RAG and computer vision. Keen to combine strong backend engineering foundations with AI/ML to build reliable, production-ready intelligent systems.

TECHNICAL SKILLS

- **Languages:** Python, Ruby, Java, SQL, JavaScript
- **Frameworks:** Ruby on Rails, Spring/Spring Boot, Flask
- **Databases & Persistence:** PostgreSQL, MySQL, MongoDB, Redis (Caching), Database Design, Query Optimisation, Object-Relational Mapping (ORM), Active Record, Hibernate, JPA, JDBC
- **Architecture & Design:** Microservices, Event-Driven Architecture, Distributed Systems, RESTful APIs (JSON), MVC, OOP/OOD, SOLID, Design Patterns
- **Messaging & Async:** Apache Kafka, SNS, SQS, Sidekiq, Karafka
- **Cloud, DevOps & Observability:** AWS (EC2, S3, RDS, DMS, MSK, CloudWatch), Docker, Kubernetes, Jenkins, CI/CD, Linux, Bash, Sentry, New Relic, Kibana
- **Testing:** JUnit, Mockito, RSpec, TDD, Unit Testing, Integration Testing
- **Tools & Collaboration:** Git, Bitbucket, Postman, Swagger/OpenAPI, Jira, Confluence, Agile/Scrum
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, NLP, Computer Vision, Generative AI, LLMs, VLMs, RAG, Embeddings, Vector Databases (FAISS, pgvector), Fine-tuning (PEFT/LoRA), Model Evaluation

PROFESSIONAL EXPERIENCE

Software Development Engineer

Oct 2022 - Jun 2025

Tata 1mg - India's Leading E-pharmacy & Digital Health Platform

Bangalore, India

- Built and maintained backend services for ODIN, Tata 1mg's in-house warehouse and supply chain management platform, using **Ruby on Rails**, **Python**, **PostgreSQL** and **AWS** microservices to power inventory, order fulfilment and procurement across multiple warehouses and retail stores.
- Cut P99 latency of high-traffic APIs by 75% by optimising **ActiveRecord** queries through eager loading, database indexing and reducing unnecessary database calls, and replaced row-by-row writes with bulk **SQL** operations in goods-return workflows.
- Improved inventory reconciliation accuracy by 25%+ across 50+ vendors by processing vendor updates in order with **AWS SQS FIFO** and vendor-specific message groups, while tracking failed events in **Redis** for retry and monitoring.
- Automated processing of 500+ B2B purchase orders a month by integrating ODIN with internal finance systems and SAP through **RESTful APIs**, and built automatic order expiry and status updates using **Sidekiq** and **Apache Kafka**, keeping downstream services in sync through **event-driven** messaging.

- Built a centralised audit-logging pipeline using **AWS DMS**, **MSK** and **Karafka** to capture **CDC events** across multiple database tables and maintain an audit history with change and user metadata.
- Built backend automation for direct Picker-to-Packer handover, removing a manual fulfilment step and saving Rs. 6 lakhs/month (approx. £4.6K); contributed to the company-wide SKU barcoding rollout, which reduced manpower costs by Rs. 20 lakhs (approx. £15.5K).
- Delivered high-priority features in parallel with on-call production support, resolving 15+ incidents a month using **Sentry**, **New Relic** and **Kibana** with warehouse operations teams, while contributing to technical design discussions, documenting solutions in **Confluence** and mentoring 3+ junior engineers.

Software Engineer

Jul 2021 - Oct 2022

Tata Digital Health - Telemedicine Platform

Bangalore, India

- Built backend **RESTful APIs** using **Java**, **Spring Boot**, **Spring Data JPA/Hibernate** and **MySQL** within a **microservices** architecture, supporting patient consultations and digital prescription (Rx) issuance and retrieval for 1,000+ consultations per day.
- Developed Clinical Decision Support System (CDSS) features, including demographic-based symptom prompts and language-based doctor matching across 4+ languages, reducing consultation setup time by 15%.
- Integrated **Single Sign-On (SSO)** and appointment-reminder push notifications, and contributed to the paid e-consultation cart-to-order and payment flow, covering key stages from authentication and booking through payment.

Software Engineering Intern

Feb 2021 - Jun 2021

Cognizant - Global Technology Services & Consulting Company

India

- Developed a full-stack application end to end, building the backend in **Java** and **Spring Boot** with a layered **MVC** design and integrating it with the frontend, applying **OOP**, **SOLID** and **TDD** with **JUnit** and **Mockito** to deliver modular, well-tested features.
- Led a 4-member team through **Agile/Scrum** sprints, owning task planning, stand-ups and delivery, and was selected for a return offer as Programmer Analyst Trainee based on performance.

PROJECTS

NutriVision - AI Dietary Assessment

- Fine-tuned **BLIP-2**, a multimodal vision-language model (**VLM**), for automated dietary assessment from food images using **LoRA** with **PyTorch**, **Hugging Face Transformers** and **PEFT**.
- Benchmarked three LoRA fine-tuning configurations across five tasks on the Nutrition5K dataset, running a structured **model evaluation** to compare accuracy and efficiency trade-offs.
- Designed a zero-cost inference-time post-processing pipeline that improved dietary reasoning accuracy without additional training or parameters, and built a **Gradio** demo to showcase results.

Healthcare Monitoring Using Machine Learning - Published, IJPRSE

- Built a **Python**, **Flask** and **scikit-learn** web app predicting patient health-risk from biometric inputs.
- Trained a Decision Tree classifier with **Pandas** and **NumPy** for data cleaning and preprocessing, serialised the model with **joblib**, and served real-time predictions through a **Bootstrap** web interface.

EDUCATION

MSc Advanced Computer Science

Sep 2025 - Sep 2026

University of Manchester - Global Future Scholarship Recipient

Manchester, United Kingdom

BTech in Information Technology

Aug 2017 - Apr 2021

Sona College of Technology - CGPA: 8.92 / 10

Tamil Nadu, India